DESCRIPTION OF COURSEWORK

Course Code	EEE302
Course Name	Electromagnetic Field Theory and Waves
Lecturer	Associate Prof. Dr. Zulkifly Abbas
Academic Session	2024/09
Assessment Title	Lab report: Determination of Electrical Specifications of a Coaxial Cable Using a Network Analyzer

A. Introduction/ Situation/ Background Information

Coaxial cable is a radio frequency (RF) transmission line to transport RF signals with relatively low loss from one point to another. The properties of standard coaxial type cables have been standardized since several decades ago. Standardization makes the selection of cables much easier as the engineers do not need to depend on any single manufacturer for cables for a particular standard from different manufacturers.

Assignments for this coursework:

Determination of Electrical Specifications of a Coaxial Cable Using a Network Analyzer (lab report)

The most important characteristics of the coaxial cable which are measurable using the N9913A FieldFox Network Analyzer shall be determined. This assignment is designed for students to apply electromagnetic theory to determine the electrical specifications of coaxial cable using the N9923A FieldFox Analyzer.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 3 Measure/determine the characteristic of transmission lines using instruments and software or other specified tools

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All the assessment submitted must be the outcome of the student. Any form of academic misconduct is a serious offense which will be penalised by being given a zero mark for the entire assessment in question or part of the assessment in question. If there is more than one guilty party as in the case of collusion, both you and your collusion partner(s) will be subjected to the same penalty.

D. Instruction to Students

The report must be submitted (softcopy/ hardcopy) before 14 July 2024. The reports must include abstract ,introduction, theory, methodology, results &discussion, conclusion and references.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Determination of Electrical Specifications of a Coaxial Cable Using a Network Analyzer	20%
	TOTAL	

F. Tasks

1. Measure the S-parameters of the cable S_{11},S_{21}, $S_{22,}$ S_{12} and saved the data in either ASCII or S2p format
2. Plot the graph attenuation vs. frequency Attenuation
3. For any chosen frequency, determine
   - VSWR
   - Impedance
   - Group delay
   - Attenuation constant
4. Apply the theory electromagnetic wave propagation to analyse all the measurement results obtained using the Fieldfox analyzer

COMPLEX ENGINEERING PROBLEM

Description of Complex Engineering Problem	
Project Theme: Determination of Electrical Specifications of a Coaxial Cable Using a Network Analyzer **Assessment: Presentation/Report/Demonstration**	
Attributes	**Remarks**
WP1 Depth of knowledge required Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamental-based, first principles analytical approach.	Requires students to analyse Fieldfox microwave measurement results using the theory of electromagnetic waves **(WK6)**
WP2 Range of conflicting requirements Involve wide-ranging or conflicting technical, engineering and other issues.	Students will need to distinguish between 1. S21, attenuation and attenuation constant 2. attenuation constant and phase constant 3. S11, reflection coefficient and VSWR 4. Phase, phase constant and group delay 5. Characteristic impedance and load impedance 6. Resistance and Reactance using the Smith Chart
WP3 Depth of analysis required Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models	Students need to use theory of electromagnetic waves and transmission line theory to determine the relationship 1.between capacitance, inductance, absolute impedance with the measured S_{11} 2.complex propagation constant, attenuation constant and phase constant

WP7 Interdependence Are high level problems including many component parts or sub-problems	Determination of the Coaxial cable electrical specifications using network analyzer is a high level problem which have different components: the theory of wave propagation, transmission line theory, the network analyzer skills required for the measurement procedure, results and analysis.

No.	EAC PO Attribute	Description	Bloom's Taxonomy Domains	WK	WP/ EA
PLO5 / CLO3	**Modern Tool Usage**	Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering activities, with an understanding of the limitations	Psychomotor	WK6	WP

APPENDIX 1

MARKING RUBRICS

	Excellent	Good	Adequate	Poor	Lacking
Abstract/ Introduction/ Theory	1.Includes complete well-written information on the background to understand the subject of the experiment and its significance. 2.Summarizes what is already known on the subject. 3. Includes the question to be answered by the lab and states hypothesis that is based on research and/or sound reasoning, described in the background. 4. Includes the description on how the hypothesis is going to be tested. 5. Deep understanding of the subject as well as creativity is demonstrated.	1.Includes strong introductory information, most major points provided. 2.Most important details in the summary provided. 3. Most important details of the hypothesis covered. 4. An idea on how to test the hypothesis is included. 5. Good understanding is demonstrated.	1.Includes good introductory information, missing some points in the summary. 2.Hypothesis is missing some minor details. 3.No description on how the hypothesis is going to be tested is provided. 4. Appropriate understanding of the subject is demonstrated.	1. Introductory information provided, missing some major points. 2. Important details of the hypothesis are missing. 3. No description on how the hypothesis is going to be tested provided. 4. Limited understanding of the subject is demonstrated.	1. Part is missing or plagiarized. OR 2.Little background information is present. 3.Incorrect information is provided. 4. For the hypothesis, important experimental details are missing, of the hypothesis is missing. 5. Poor understanding or no understanding of the subject is demonstrated.
methodology	Description (not a copy of a protocol!) is included, detailed enough so it could be repeated by another scientist. Description is not overloaded with unnecessary details.	Description included, some steps are vague or unclear OR instead of description, a step-by–step protocol is copied from manual.	The description gives generalities, enough for reader to understand how the experiment was conducted	Would be impossible to repeat the experiment, reader must guess at how the data was gathered or experiment conducted	Part is missing or plagiarized

Results	Results and data are clearly recorded, so it is easy for the reader to see trends. Excellent graphs, tables or figures with appropriate names and labels, excellent information and understanding of results, excellent interpretation of data and trends while referring to figures and tables.	Results are clear, but trends are not obvious or there are minor errors in organization. Good figures, graphs, or tables, good information results, good understanding of results, but some figure/table names or labels are missing.	Results are unclear, trends are not obvious, disorganized, but there is enough data to show the experiment was conducted. Appropriate figures, graphs or tables or drawings, some labels and figure names are missing. Only some understanding of results is evident.	Results are disorganized or poorly recorded, do not make sense; not enough data was taken to justify results. Missing or incorrect information.	Part is missing or plagiarized. Misleading, confusing and/or incorrect information. Poorly presented results.
Discussion/ Conclusions	1.Summarizes data used to draw conclusions 2.Conclusions follow data (not wild guesses or leaps of logic) 3. Hypothesis is rejected or accepted based on the data. 4. Discussion of applications or real world connections is present. 5. Deep understanding of the subject as well as creativity is demonstrated.	Good conclusion, important information provided, or one point from Excellent is missing	Appropriate conclusion. Two points from Excellent are missing.	Limited conclusion. Important information is missing.	Part is missing or plagiarized. OR No conclusion is made. Missing important information. Little effort or reflection is shown.
References	All references and citations are correctly written and present. More than two resources are cited. Student accurately cites to the sources throughout the text.	One reference or citation is missing or incorrectly present.	Appropriate citing and reference list.	References and citations are missing or errors detract significantly from paper.	Part is missing or plagiarized. Incorrect citing.

APPENDIX 1

MARKING RUBRICS

Component Title	ASSIGNMENT				Percentage (%)	20	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
Abstract						10%	2
Introduction & Theory						10%	2
Methodology/ Experimental						40%	8
Results, Disscussion Conclusion, Formatting References						40%	8
					TOTAL	100%	20

Criteria	Score and Descriptors				
	Excellent (5)	**Good (4)**	**Average (3)**	**Need Improvement (2)**	**Poor (1)**
Abstract	Clear, concise, and thorough summary including context, important results, and conclusions.	Refers to most of the major points, but some minor details are missing or not clearly explained	Misses one of more major parts of the results, context, or conclusions	Missing several majors aspects and merely repeats information from the introduction	None or unrelated
Introduction	A cohesive, well-written summary (including all relevant chemistry) of the background material pertinent to the experiment with appropriate references. Places the purpose of the experiment in context.	Is nearly complete but does not provide context for minor points. Contains relevant information but fails to provide background for one aspect of the experiment, or certain information is not cohesive.	Certain major introductory points are missing (ex: background, theory, chemistry, context, etc.) or explanations are unclear and confusing. References are used properly.	Very little background information is provided and/or information is incorrect. No references are provided.	None or unrelated
Experimental	Contains details on how the experiment was performed and the procedures followed. Written in the correct tense and omits information that can be assumed by peers (trained chemists)	Narrative includes most important experimental details but is missing one or more relevant pieces of information.	Missing several experimental details or some incorrect statements.	Several important experimental details are missing. Narrative is incorrect, illogical, or copied directly from the lab manual. Written in the incorrect tense.	None or unrelated
Results (Presentation of results, figures and tables)	All figures, graphs, and tables are numbered with appropriate captions. All tables, figures, etc. are explicitly mentioned in the text. Relevant experimental data are presented which are used in the discussion.	All figures, graphs, and tables are correctly drawn, but some have minor problems that could be still be improved. All data and associated figures, etc. are mentioned in the text. Most relevant data present.	Most figures, graphs, and tables are included, but some important or required features are missing. Certain data reported are not mentioned in the text or are missing. Captions are not descriptive or incomplete.	Figures, graphs, and tables are poorly constructed; have missing titles, captions or numbers. Certain data reported are not mentioned in the text. Important data missing.	None or unrelated